

LM Physics of Climate

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2 How do pressure and temperature vary with height?

In a non-escaping atmosphere, pressure is determined by hydrostatic equilibrium (Section 2.1). The vertical temperature profile depends on the dominant energy transport mechanism. In the next few lectures, we will look at two key modes of vertical energy transport in planetary atmospheres: radiation and convection. Sections 2.2 and 2.3 go through some of the thermodynamics we will need to start tackling convection in Section 2.4.

2.1 Introducing pressure: hydrostatic equilibrium

How does the pressure in an atmosphere vary with altitude? For the case of a stable, non-escaping atmosphere, we can work this out by balancing the forces on a parcel of gas.

Consider the cylinder of gas in Figure 1, with cross-sectional surface area A , density ρ and height δz . The pressure at the base of the parcel is P , while at the top the pressure is $P + \delta P$. In a hydrostatic atmosphere, this imbalance of pressure forces is compensated by the gravitational force. Note that in an atmosphere, δP is negative.

Balancing the upwards and downwards forces on the parcel of gas, we obtain

$$\frac{dP}{dz} = -\rho g. \quad (2.1)$$

To solve this equation, we can use the ideal gas law to write ρ in terms of P and T : $P = \frac{k_B}{\mu} \rho T$, where ρ is the mass density of the gas and μ is its mean molecular weight (e.g. for pure H_2 gas $\mu = 2 \text{ u} = 3.32 \times 10^{-27} \text{ kg}$). This substitution gives:

$$\frac{d \ln P}{dz} = -\frac{\mu g}{k_B T}. \quad (2.2)$$

In the simple case of an isothermal atmosphere, the solution to Equation 2.2 is

$$P = P_0 e^{-z/H_{sc}}, \quad (2.3)$$

where the scale height of the atmosphere,

$$H_{sc} = \frac{k_B T}{\mu g}. \quad (2.4)$$

The scale height is the distance over which pressure decreases by a factor of e , and gives a sense of the vertical extent of an atmosphere. For example, an atmosphere which is hotter, has a lower mean molecular weight and/or a lower gravity has a ‘puffier’ atmosphere.

We can estimate H_{sc} for the Earth using its surface temperature, $T_s = 288 \text{ K}$, the mean molecular weight of dry air, $\mu = 28.97 \text{ u}$, and $g = 9.81 \text{ m/s}^2$. This gives us $H_{sc} = 8.4 \text{ km}$, or close to the height of Mount Everest (8.85 km).

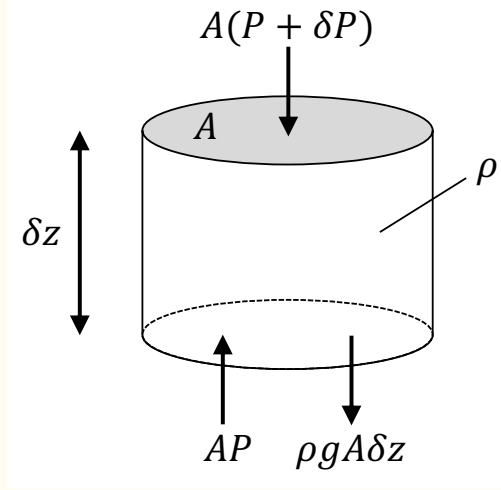


Figure 1: The pressure and gravitational forces acting on a parcel of gas with density ρ , height δz , cross-sectional surface area A and gravitational acceleration g .

For Venus, the surface temperature is ~ 740 K, $\mu \approx 43.45$ u, and $g = 8.87 \text{ m/s}^2$, which gives $H_{sc} = 15.9$ km. For Mars, the surface temperature is ~ 214 K, $\mu \approx 43.49$ u, and $g = 3.72 \text{ m/s}^2$, which gives $H_{sc} = 11.0$ km.

Figure 2 shows how pressure varies with altitude for the Earth, Venus and Mars, using the isothermal approximation and Equation 2.3. Venus has a much thicker atmosphere than Earth or Mars, resulting in a much higher surface pressure.

2.2 The ideal gas law

There are several ways of writing the ideal gas law, each using different constants and variables. It is worth laying these out side by side to avoid confusion and to aid in converting between them. In this course, I will predominantly use the following forms:

1. In terms of number density,

$$P = nk_{\text{B}}T, \quad (2.5)$$

where P is pressure, n is the number density of particles, k_{B} is Boltzmann's constant and T is temperature.

2. In terms of mass density,

$$P = \frac{k_{\text{B}}}{\mu} \rho T, \quad (2.6)$$

where $\rho = \mu n$ is the mass density of the gas and μ is its mean molecular weight (e.g. for pure H_2 gas $\mu = 2$ u = 3.32×10^{-27} kg).

3. in terms of volume and number of particles,

$$PV = Nk_{\text{B}}T, \quad (2.7)$$

where V is the volume of gas and N is the total number of particles.

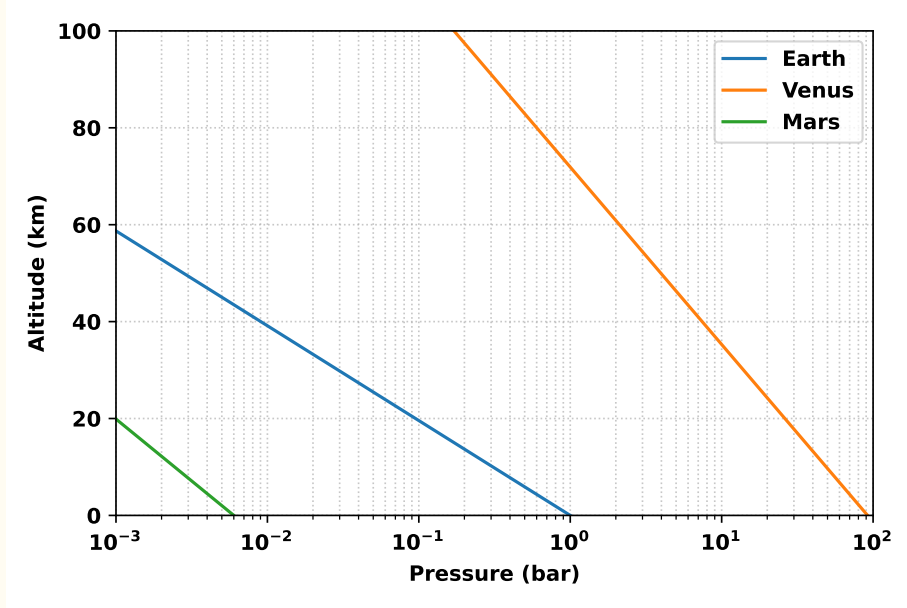


Figure 2: Profiles of altitude vs pressure for the Earth, Venus and Mars, using the isothermal solution to the equation of hydrostatic equilibrium.

4. We will sometimes use proportionalities between P , n , ρ , V and T to link their derivatives;

$$P \propto nT \implies d \ln P = d \ln n + d \ln T, \quad (2.8)$$

$$P \propto \rho T \implies d \ln P = d \ln \rho + d \ln T, \quad (2.9)$$

$$PV \propto T \implies d \ln P + d \ln V = d \ln T. \quad (2.10)$$

It is also common to see the ideal gas law written in terms of the ideal gas constant, $R = N_A k_B$. N_A is Avogadro's constant, the number of molecules in one mole of a substance. The ideal gas law then becomes

$$PV = N_{\text{mol}} RT, \quad (2.11)$$

where N_{mol} is the number of moles of gas.

You may also come across the specific gas constant,

$$R_{\text{specific}} = \frac{R}{M_{\text{molar}}} = \frac{N_A k_B}{N_A \mu} = \frac{k_B}{\mu},$$

where M_{molar} is the molar mass of the gas.

2.3 Some useful thermodynamics

Note: none of the derivations in Section 2.3 are examinable. We will be using the resulting formulae when studying convection.

Convection involves parcels of gas rising and expanding adiabatically. Therefore, it is useful to begin with the thermodynamics of an adiabatic parcel. Let's begin with the first law of thermodynamics:

$$dU = dQ + dW, \quad (2.12)$$

where U is the internal energy of the gas, dQ is the heat supplied to the gas, and dW is the work done on the system.

2.3.1 Heat capacity and Mayer's relation

The heat capacity of a gas describes the energy needed to raise its temperature by 1 K. C_V and C_P are the heat capacities at constant volume and pressure, respectively, and have SI units of J/K. It can be shown that for an ideal gas, C_V and C_P are related by Mayer's relation:

$$C_P - C_V = N_{\text{mol}}R. \quad (2.13)$$

It is also useful to define the specific heat capacities at constant pressure and volume, respectively: $c_P = C_P/M$ and $c_V = C_V/M$ (both with SI units of J/(kg·K)), where M is the total mass of gas. Mayer's relation can be rewritten in terms of the specific heat capacities as:

$$c_P - c_V = \frac{N_{\text{mol}}R}{M} = \frac{k_B}{\mu}, \quad (2.14)$$

where μ is the mean molecular weight of the gas.

2.3.2 Adiabatic gases

For an adiabatic process, $dQ = 0$. Furthermore, for an ideal gas, the internal energy can be completely described by temperature, T , and

$$dU = C_V dT. \quad (2.15)$$

For a reversible change, we also have

$$dW = -PdV. \quad (2.16)$$

Substituting this into Equation 2.12 for a single mole of gas gives

$$\begin{aligned} C_V dT &= -PdV \\ &= -\frac{N_{\text{mol}}RT}{V}dV, \end{aligned} \quad (2.17)$$

where the ideal gas law has been used to rewrite P in the second line and N_{mol} is the number of moles of gas. Using the definition $\gamma = C_P/C_V$ and Mayer's relation (Equation 2.13), we can rewrite Equation 2.17 as

$$\begin{aligned} d \ln T &= -\frac{N_{\text{mol}}R}{C_V} d \ln V \\ &= (1 - \gamma) d \ln V. \end{aligned}$$

Integrating this, and using the fact that for an ideal gas $PV \propto T$, we get the following equivalent relations:

$$TV^{\gamma-1} = \text{constant}, \quad (2.18)$$

$$P^{1-\gamma}T^\gamma = \text{constant}, \quad (2.19)$$

$$PV^\gamma = \text{constant}. \quad (2.20)$$

2.3.3 Heat supplied to an ideal gas

When we later consider an adiabatic gas with condensable components, it will be useful to express the heat supplied to an ideal gas in terms of its heat capacities. We will ultimately set $dQ = 0$ for an adiabatic system.

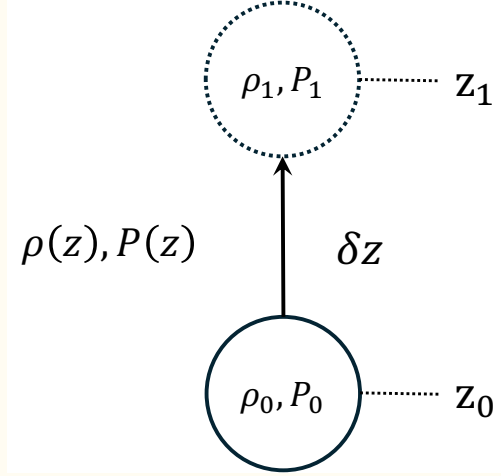


Figure 3: Schematic of a gas parcel rising by a distance δz in a background medium with density and pressure $\rho(z)$ and $P(z)$, respectively.

Following on from the first law (Equation 2.12) and Equations 2.15 and 2.16,

$$dQ = C_V dT + P dV \quad (2.21)$$

$$\begin{aligned} &= C_V dT + d(PV) - V dP \\ &= C_V dT + N k_B dT - V dP \\ &= M \left(c_V dT + \frac{k_B}{\mu} dT - \frac{dP}{\rho} \right) \\ &= M \left(c_P dT - \frac{dP}{\rho} \right), \end{aligned} \quad (2.22)$$

where we have used the ideal gas law in the third line and factorised the total mass of gas, M .

2.4 Dry convection

We will begin by considering ‘dry’ convection, i.e. where no condensation occurs. Consider a parcel of gas with initial density ρ_0 and initial pressure P_0 and located at height z_0 . It is surrounded by a medium of density $\rho(z)$ and pressure $P(z)$. As the parcel moves upwards by a small distance, δz , it expands adiabatically and its density and pressure become ρ_1 and P_1 , respectively (Figure 3).

Since pressure equilibrates quickly, we can say that $P_0 = P(z_0)$ and $P_1 = P(z_0 + \delta z)$. The parcel of gas expands adiabatically, so we also know that

$$P_0 \rho_0^{-\gamma} = P_1 \rho_1^{-\gamma} \quad (2.23)$$

(this follows from Equation 2.20 and the fact that $\rho \propto 1/V$).

What we ultimately want to work out is: after this small vertical motion, will the parcel of gas be buoyant and continue to rise, or will it sink again? If it sinks, the gas will be stable to convection. For the parcel to sink, we require:

$$\rho_1 > \rho(z_0 + \delta z). \quad (2.24)$$

The reason this can occur is that the density of the parcel is determined by *adiabatic expansion* (Equation 2.23), whereas the density of the background gas follows the *ideal gas law*, $P \propto \rho T$. The rate of change of the background density with height therefore depends on its vertical temperature and pressure gradients, which in turn determine whether the gas is stable or unstable to convection.

Rearranging Equation 2.23, we find that the left-hand side of Equation 2.24 is

$$\begin{aligned} \rho_1 &= \rho_0 \left(\frac{P_1}{P_0} \right)^{1/\gamma} \\ &= \rho(z_0) \left(\frac{P(z_0 + \delta z)}{P(z_0)} \right)^{1/\gamma} \\ &= \rho(z_0) \left(1 + \frac{1}{P} \frac{dP}{dz} \delta z \right)^{1/\gamma} \\ &\approx \rho(z_0) \left(1 + \frac{1}{\gamma P} \frac{dP}{dz} \delta z \right), \end{aligned} \quad (2.25)$$

where in the penultimate line we have used the fact that $P(z_0 + \delta z) = P(z_0) + \frac{dP}{dz} \delta z$, and in the last line we have used the fact that δz is small to do a Taylor series expansion.

The right-hand side of Equation 2.24 can also be expanded as a Taylor series:

$$\rho(z_0 + \delta z) = \rho(z_0) + \frac{d\rho}{dz} \delta z.$$

Since the background gas follows the ideal gas law, we can use the fact that $\ln P = \ln \rho + \ln T + \text{constant}$, and hence $\frac{d \ln P}{dz} = \frac{d \ln \rho}{dz} + \frac{d \ln T}{dz}$. Substituting into the above, we get:

$$\begin{aligned} \rho(z_0 + \delta z) &= \rho(z_0) + \rho \left(\frac{d \ln P}{dz} - \frac{d \ln T}{dz} \right) \delta z \\ &\approx \rho(z_0) + \rho(z_0) \left(\frac{d \ln P}{dz} - \frac{d \ln T}{dz} \right) \delta z, \end{aligned} \quad (2.26)$$

where the last line follows from the fact that we are only expanding to first order in δz .

Substituting Equations 2.25 and 2.26 into Equation 2.24, we get:

$$\frac{\rho(z_0)}{\gamma P} \frac{dP}{dz} > \rho(z_0) \left(\frac{d \ln P}{dz} - \frac{d \ln T}{dz} \right).$$

Note that both sides of this inequality are negative, which is why the inequality sign from Equation 2.24 has flipped. After some rearranging, this becomes:

$$\frac{d \ln T}{dz} > \left(1 - \frac{1}{\gamma} \right) \frac{d \ln P}{dz}. \quad (2.27)$$

This is the Schwarzschild criterion for convective stability. We can rewrite this again as a condition on the temperature-pressure gradient of the gas:

$$\frac{d \ln T}{d \ln P} < \frac{\gamma - 1}{\gamma}. \quad (2.28)$$

This is commonly written as

$$\nabla < \nabla_{\text{ad}}, \quad (2.29)$$

where $\nabla \equiv \frac{d \ln T}{d \ln P}$ and $\nabla_{\text{ad}} \equiv \frac{\gamma-1}{\gamma}$.

2.5 The dry lapse rate

In the context of Earth's climate, it is common to write the Schwarzschild criterion in terms of the *lapse rate*, $-\frac{dT}{dz}$. To do this, we will assume that the gas is in hydrostatic equilibrium (a good assumption for the Earth's atmosphere), and use some thermodynamics. Combining $\frac{dP}{dz} = -\rho g$, $P = nk_{\text{B}}T$ and $\rho = n\mu$ we get

$$T \frac{dP}{P} = -\frac{\mu g}{k_{\text{B}}} dz. \quad (2.30)$$

For an adiabatic gas, we can also differentiate Equation 2.19 to get

$$\gamma \frac{dT}{T} = (\gamma - 1) \frac{dP}{P}. \quad (2.31)$$

Substituting Equation 2.31 into 2.30 gives

$$\begin{aligned} -\frac{dT}{dz} &= \frac{\gamma - 1}{\gamma} \frac{\mu g}{k_{\text{B}}} \\ &= \frac{N_{\text{mol}} R}{C_p} \frac{\mu g}{k_{\text{B}}} \\ &= \frac{g}{c_p}, \end{aligned} \quad (2.32)$$

noting that from Section 2.3 $(\gamma - 1)/\gamma = N_{\text{mol}} R / C_p$, and that $R = N_{\text{A}} k_{\text{B}}$. c_p is the specific heat capacity at constant pressure, i.e. the specific heat capacity at constant pressure per unit mass. Equation 2.32 is the dry lapse rate of an atmosphere.